SG No. 74 D_{2h}^{28} *Imma* [orthorhombic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `4a`, site symmetry: `2/m..`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-4,-5,8]`
2	$[0,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-3,-6,7]`

Table 2: Wyckoff bond: `4b@4a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[0,\ 0,\ 0]$	`[1,4,-5,-8]`
2	$[-X,\ 0,\ 0]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,3,-6,-7]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ 0]$	`[1,-5]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,-6]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,-7]`
4	$[X,\ -Y,\ -Z]$	$[0,\ 0,\ 0]$	`[4,-8]`

* Wyckoff site: `4b`, site symmetry: `2/m..`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-4,-5,8]`
2	$[0,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,-3,-6,7]`

Table 5: Wyckoff bond: `4b@4b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,4,-5,-8]`
2	$[-X,\ 0,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,-6,-7]`

Table 6: Wyckoff bond: `8c@4b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,-5]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,-6]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,-7]`

4	$[X, -Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `4c`, site symmetry: `.2/m.`

Table 7: Wyckoff bond: `4a@4c`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,-4,-6,8]`

Table 8: Wyckoff bond: `4b@4c`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,3,-5,-7]`
2	$[0, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,4,-6,-8]`

Table 9: Wyckoff bond: `8c@4c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,-8]`

* Wyckoff site: `4d`, site symmetry: `.2/m.`

Table 10: Wyckoff bond: `4a@4d`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-3,-5,7]`
2	$[-X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-4,-6,8]`

Table 11: Wyckoff bond: `4b@4d`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,3,-5,-7]`
2	$[0, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,4,-6,-8]`

Table 12: Wyckoff bond: `8c@4d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,-7]`

continued ...

Table 12

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,-8]`

* Wyckoff site: `4e`, site symmetry: `mm2`

Table 13: Wyckoff bond: `4a@4e`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,2,7,8]`
2	$[0, 0, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[3,4,5,6]`

Table 14: Wyckoff bond: `4b@4e`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2,-7,8]`
2	$[0, Y, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[3,-4,-5,6]`

Table 15: Wyckoff bond: `4c@4e`, bond symmetry: `mm2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2,7,-8]`
2	$[-X, 0, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[3,-4,5,-6]`

Table 16: Wyckoff bond: `8d@4e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[3,-4]`
3	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[5,-6]`
4	$[X, -Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[7,-8]`

Table 17: Wyckoff bond: `8e@4e`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,7]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2,8]`
3	$[-X, 0, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[3,5]`
4	$[X, 0, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[4,6]`

Table 18: Wyckoff bond: `8f@4e`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[4,5]`

Table 19: Wyckoff bond: `16g@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[0, \frac{3}{4}, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[8]`

* Wyckoff site: `8f`, site symmetry: `2..`

Table 20: Wyckoff bond: `8a@8f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,-3]`
3	$[0, -Y, -Z]$	$[-x, 0, 0]$	`[5,-8]`
4	$[0, Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[6,-7]`

Table 21: Wyckoff bond: `8b@8f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,3]`
3	$[-X, 0, 0]$	$[-x, 0, 0]$	`[5,8]`
4	$[X, 0, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[6,7]`

Table 22: Wyckoff bond: `16c@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[-X, -Y, -Z]$	$[-x, 0, 0]$	`[5]`

continued ...

Table 22

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	`[7]`
8	$[-X, Y, Z]$	$[-x, 0, 0]$	`[8]`

* Wyckoff site: `8g`, site symmetry: `.2.`

Table 23: Wyckoff bond: `8a@8g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{2} - y, \frac{1}{4}\right]$	`[2,-4]`
3	$[-X, 0, -Z]$	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	`[5,-7]`
4	$[X, 0, -Z]$	$\left[\frac{1}{4}, y + \frac{1}{2}, \frac{3}{4}\right]$	`[6,-8]`

Table 24: Wyckoff bond: `8b@8g`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	`[1,3]`
2	$[0, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{2} - y, \frac{1}{4}\right]$	`[2,4]`
3	$[0, -Y, 0]$	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	`[5,7]`
4	$[0, Y, 0]$	$\left[\frac{1}{4}, y + \frac{1}{2}, \frac{3}{4}\right]$	`[6,8]`

Table 25: Wyckoff bond: `16c@8g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{2} - y, \frac{1}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{2} - y, \frac{1}{4}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{1}{4}, y + \frac{1}{2}, \frac{3}{4}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{1}{4}, y + \frac{1}{2}, \frac{3}{4}\right]$	`[8]`

* Wyckoff site: `8h`, site symmetry: `m..`

Table 26: Wyckoff bond: `8a@8h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, Y, Z]$	$[0, y, z]$	`[1,8]`
2	$[0, -Y, Z]$	$\left[0, \frac{1}{2} - y, z\right]$	`[2,7]`
3	$[0, Y, -Z]$	$\left[0, y + \frac{1}{2}, -z\right]$	`[3,6]`
4	$[0, -Y, -Z]$	$[0, -y, -z]$	`[4,5]`

Table 27: Wyckoff bond: `8b@8h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	`[1,-8]`
2	$[-X, 0, 0]$	$\left[0, \frac{1}{2} - y, z\right]$	`[2,-7]`
3	$[-X, 0, 0]$	$\left[0, y + \frac{1}{2}, -z\right]$	`[3,-6]`
4	$[X, 0, 0]$	$[0, -y, -z]$	`[4,-5]`

Table 28: Wyckoff bond: `16c@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2} - y, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, y + \frac{1}{2}, -z\right]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -z]$	`[4]`
5	$[-X, -Y, -Z]$	$[0, -y, -z]$	`[5]`
6	$[X, Y, -Z]$	$\left[0, y + \frac{1}{2}, -z\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[0, \frac{1}{2} - y, z\right]$	`[7]`
8	$[-X, Y, Z]$	$[0, y, z]$	`[8]`

* Wyckoff site: `8i`, site symmetry: `.m.`

Table 29: Wyckoff bond: `8a@8i`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[x, \frac{1}{4}, z\right]$	`[1,7]`
2	$[-X, 0, Z]$	$\left[-x, \frac{1}{4}, z\right]$	`[2,8]`
3	$[-X, 0, -Z]$	$\left[-x, \frac{3}{4}, -z\right]$	`[3,5]`
4	$[X, 0, -Z]$	$\left[x, \frac{3}{4}, -z\right]$	`[4,6]`

Table 30: Wyckoff bond: `8b@8i`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[x, \frac{1}{4}, z\right]$	`[1,-7]`
2	$[0, -Y, 0]$	$\left[-x, \frac{1}{4}, z\right]$	`[2,-8]`
3	$[0, Y, 0]$	$\left[-x, \frac{3}{4}, -z\right]$	`[3,-5]`
4	$[0, -Y, 0]$	$\left[x, \frac{3}{4}, -z\right]$	`[4,-6]`

Table 31: Wyckoff bond: `16c@8i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{4}, z\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, \frac{3}{4}, -z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, \frac{3}{4}, -z\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}, -z\right]$	`[5]`

continued ...

Table 31

No.	vector	center	mapping
6	$[X, Y, -Z]$	$\left[x, \frac{3}{4}, -z\right]$	[6]
7	$[X, -Y, Z]$	$\left[x, \frac{1}{4}, z\right]$	[7]
8	$[-X, Y, Z]$	$\left[-x, \frac{1}{4}, z\right]$	[8]

* Wyckoff site: 16j, site symmetry: 1

Table 32: Wyckoff bond: 16a@16j, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-y, z\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, y+\frac{1}{2}, -z\right]$	[3]
4	$[X, -Y, -Z]$	$[x, -y, -z]$	[4]
5	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[5]
6	$[X, Y, -Z]$	$\left[x, y+\frac{1}{2}, -z\right]$	[6]
7	$[X, -Y, Z]$	$\left[x, \frac{1}{2}-y, z\right]$	[7]
8	$[-X, Y, Z]$	$[-x, y, z]$	[8]